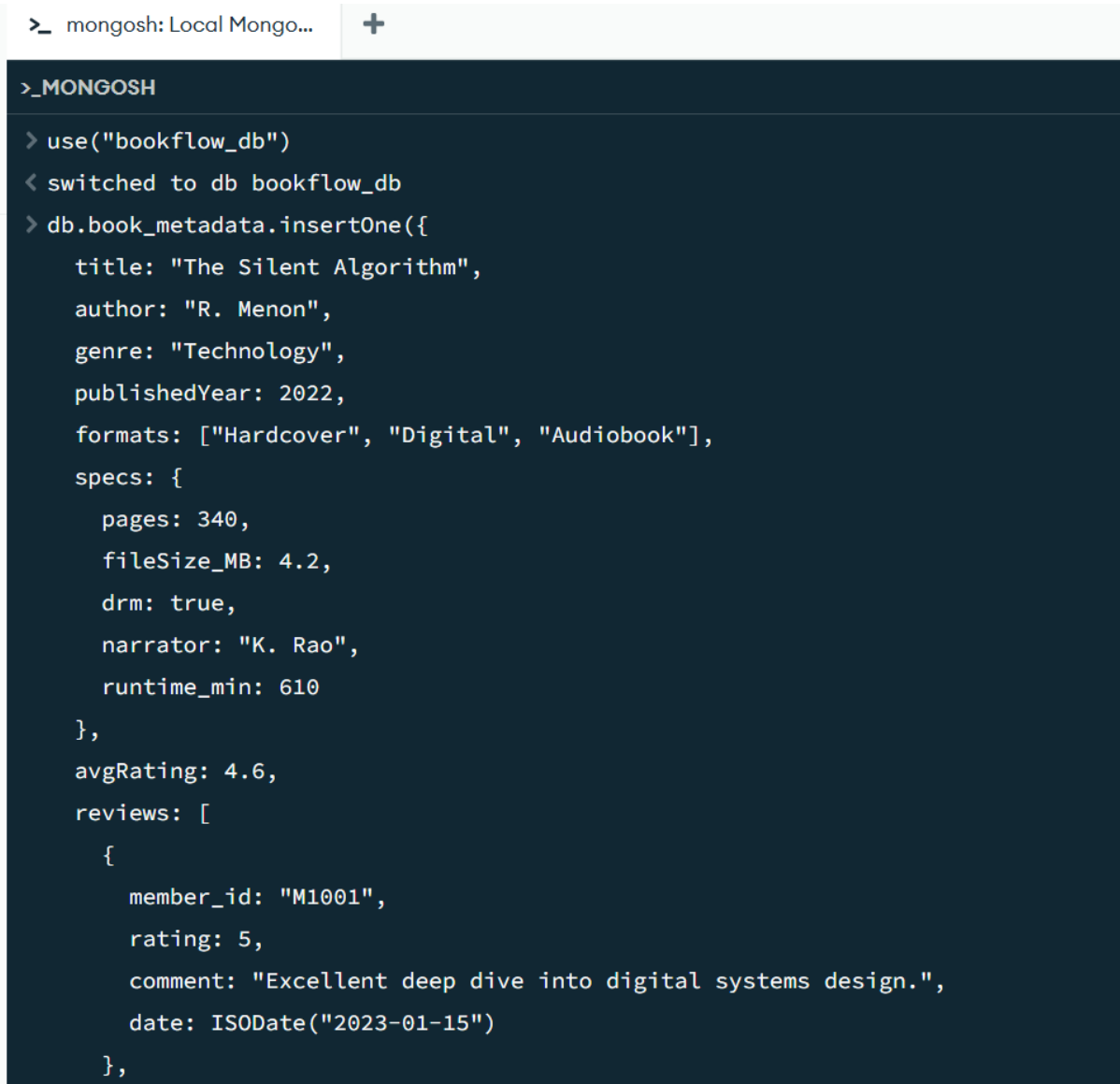


DBS – Practical3

Task 1:



```
>_ mongosh: Local Mongo... +  
  
>_MONGOSH  
  
> use("bookflow_db")  
< switched to db bookflow_db  
> db.book_metadata.insertOne({  
  title: "The Silent Algorithm",  
  author: "R. Menon",  
  genre: "Technology",  
  publishedYear: 2022,  
  formats: ["Hardcover", "Digital", "Audiobook"],  
  specs: {  
    pages: 340,  
    fileSize_MB: 4.2,  
    drm: true,  
    narrator: "K. Rao",  
    runtime_min: 610  
  },  
  avgRating: 4.6,  
  reviews: [  
    {  
      member_id: "M1001",  
      rating: 5,  
      comment: "Excellent deep dive into digital systems design.",  
      date: ISODate("2023-01-15")  
    },  
  ],  
})
```

>_MONGOSH

```
{
  member_id: "M1042",
  rating: 4,
  comment: "Good but the Digital edition had formatting issues.",
  date: ISODate("2023-02-20")
}
]
})
< {
  acknowledged: true,
  insertedId: ObjectId('6a8ad25afb25f5a9d6acfb4')
}
```

>_MONGOSH

> db.book_metadata.find()

```
< {
  _id: ObjectId('6a8ad25a0fb25f5a9d6acfb4'),
  title: 'The Silent Algorithm',
  author: 'R. Menon',
  genre: 'Technology',
  publishedYear: 2022,
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ],
  specs: {
    pages: 340,
    fileSize_MB: 4.2,
    drm: true,
    narrator: 'K. Rao',
    runtime_min: 610
  },
}
```

```
avgRating: 4.6,  
reviews: [  
  {  
    member_id: 'M1001',  
    rating: 5,  
    comment: 'Excellent deep dive into digital systems design.',  
    date: 2023-01-15T00:00:00.000Z  
  },  
  {  
    member_id: 'M1042',  
    rating: 4,  
    comment: 'Good but the Digital edition had formatting issues.',  
    date: 2023-02-20T00:00:00.000Z  
  }  
]  
}  
bookflow_db > |
```

Task 2

>_MONGOSH

```
> db.book_metadata.find(
  { avgRating: { $gt: 4 } },
  { title: 1, avgRating: 1, _id: 0 }
);
< {
  title: 'The Silent Algorithm',
  avgRating: 4.6
}
> db.book_metadata.find(
  { formats: { $in: ["Digital", "Audiobook"] } },
  { title: 1, formats: 1, _id: 0 }
);
< {
  title: 'The Silent Algorithm',
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ]
}
> db.book_metadata.find({
  avgRating: { $gt: 4 },
  formats: { $in: ["Digital"] }
});
```

>_MONGOSH

```
< {
  _id: ObjectId('6a8ad25a0fb25f5a9d6acfb4'),
  title: 'The Silent Algorithm',
  author: 'R. Menon',
  genre: 'Technology',
  publishedYear: 2022,
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ],
  specs: {
    pages: 340,
    fileSize_MB: 4.2,
    drm: true,
    narrator: 'K. Rao',
    runtime_min: 610
  },
  avgRating: 4.6,
  reviews: [
    {
      member_id: 'M1001',
      rating: 5,
      comment: 'Excellent deep dive into digital systems design.',
      date: 2023-01-15T00:00:00.000Z
    },
  ],
}
```

```
>_MONGOSH
```

```
{
  member_id: 'M1042',
  rating: 4,
  comment: 'Good but the Digital edition had formatting issues.',
  date: 2023-02-20T00:00:00.000Z
}
]
}
> db.book_metadata.find(
  { formats: { $regex: "Digital", $options: "i" } },
  { title: 1, formats: 1 }
);
< {
  _id: ObjectId('6a8ad25a0fb25f5a9d6acfb4'),
  title: 'The Silent Algorithm',
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ]
}
> db.book_metadata.find(
  { "reviews.comment": { $regex: "Digital", $options: "i" } },
  { title: 1, "reviews.$": 1 }
);
```

```
< {  
  _id: ObjectId('6a8ad25a0fb25f5a9d6acfb4'),  
  title: 'The Silent Algorithm',  
  reviews: [  
    {  
      member_id: 'M1001',  
      rating: 5,  
      comment: 'Excellent deep dive into digital systems design.',  
      date: 2023-01-15T00:00:00.000Z  
    }  
  ]  
}  
bookflow_db > |
```


Task 3

```
>_MONGOSH

> db.book_metadata.aggregate([
  { $match: { publishedYear: { $gt: 2020 } } }
]);
< {
  _id: ObjectId('6a8ad25afb25f5a9d6acfb4'),
  title: 'The Silent Algorithm',
  author: 'R. Menon',
  genre: 'Technology',
  publishedYear: 2022,
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ],
  specs: {
    pages: 340,
    fileSize_MB: 4.2,
    drm: true,
    narrator: 'K. Rao',
    runtime_min: 610
  },
}
```

```
    avgRating: 4.6,  
    reviews: [  
      {  
        member_id: 'M1001',  
        rating: 5,  
        comment: 'Excellent deep dive into digital systems design.',  
        date: 2023-01-15T00:00:00.000Z  
      },  
      {  
        member_id: 'M1042',  
        rating: 4,  
        comment: 'Good but the Digital edition had formatting issues.',  
        date: 2023-02-20T00:00:00.000Z  
      }  
    ]  
  }  
bookflow db>
```

```
>_MONGOSH
```

```
> db.book_metadata.aggregate([
  { $match: { publishedYear: { $gt: 2020 } } },
  { $unwind: "$reviews" }
]);
< {
  _id: ObjectId('6a8ad25a0fb25f5a9d6acfb4'),
  title: 'The Silent Algorithm',
  author: 'R. Menon',
  genre: 'Technology',
  publishedYear: 2022,
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ],
  specs: {
    pages: 340,
    fileSize_MB: 4.2,
    drm: true,
    narrator: 'K. Rao',
    runtime_min: 610
  },
}
```

>_MONGOSH

```
    avgRating: 4.6,
    reviews: {
      member_id: 'M1001',
      rating: 5,
      comment: 'Excellent deep dive into digital systems design.',
      date: 2023-01-15T00:00:00.000Z
    }
  }
  {
    _id: ObjectId('6a8ad25a0fb25f5a9d6acfb4'),
    title: 'The Silent Algorithm',
    author: 'R. Menon',
    genre: 'Technology',
    publishedYear: 2022,
    formats: [
      'Hardcover',
      'Digital',
      'Audiobook'
    ],
    specs: {
      pages: 340,
      fileSize_MB: 4.2,
      drm: true,
      narrator: 'K. Rao',
      runtime_min: 610
    },
```

```
    avgRating: 4.6,  
    reviews: {  
      member_id: 'M1042',  
      rating: 4,  
      comment: 'Good but the Digital edition had formatting issues.',  
      date: 2023-02-20T00:00:00.000Z  
    }  
  }  
}  
bookflow_db> |
```

```
> db.book_metadata.aggregate([  
  { $match: { publishedYear: { $gt: 2020 } } },  
  { $unwind: "$reviews" },  
  {  
    $group: {  
      _id: "$title",  
      avgReviewRating: { $avg: "$reviews.rating" },  
      totalReviews: { $sum: 1 },  
      author: { $first: "$author" }  
    }  
  }  
]);  
< {  
  _id: 'The Silent Algorithm',  
  avgReviewRating: 4.5,  
  totalReviews: 2,  
  author: 'R. Menon'  
}  
bookflow_db> |
```

```
> db.book_metadata.aggregate([
  { $match: { publishedYear: { $gt: 2020 } } },
  { $unwind: "$reviews" },
  {
    $group: {
      _id: "$title",
      avgReviewRating: { $avg: "$reviews.rating" },
      totalReviews: { $sum: 1 },
      author: { $first: "$author" }
    }
  },
  { $sort: { avgReviewRating: -1 } }
]);
< {
  _id: 'The Silent Algorithm',
  avgReviewRating: 4.5,
  totalReviews: 2,
  author: 'R. Menon'
}
bookflow_db >
```

>_MONGOSH

```
> db.book_metadata.aggregate([
  { $match: { publishedYear: { $gt: 2020 } } },
  { $unwind: "$reviews" },
  {
    $group: {
      _id: "$title",
      avgReviewRating: { $avg: "$reviews.rating" },
      totalReviews: { $sum: 1 },
      author: { $first: "$author" }
    }
  },
  { $sort: { avgReviewRating: -1 } },
  {
    $project: {
      _id: 0,
      title: "$_id",
      author: 1,
      avgReviewRating: { $round: ["$avgReviewRating", 2] },
      totalReviews: 1
    }
  }
]);
```

```
< {
  totalReviews: 2,
  author: 'R. Menon',
  title: 'The Silent Algorithm',
  avgReviewRating: 4.5
}
bookflow_db > |
```

Task 4

```
> db.book_metadata.createIndex({ "reviews.comment": "text" });
< reviews.comment_text
> db.book_metadata.getIndexes();
< [
  { v: 2, key: { _id: 1 }, name: '_id_' },
  {
    v: 2,
    key: { _fts: 'text', _ftsx: 1 },
    name: 'reviews.comment_text',
    weights: { 'reviews.comment': 1 },
    default_language: 'english',
    language_override: 'language',
    textIndexVersion: 3
  }
]
bookflow_db> |
```

```
> db.book_metadata.find(
  { $text: { $search: "Digital" } },
  { title: 1, "reviews.comment": 1, _id: 0 }
);
< {
  title: 'The Silent Algorithm',
  reviews: [
    {
      comment: 'Excellent deep dive into digital systems design.'
    },
    {
      comment: 'Good but the Digital edition had formatting issues.'
    }
  ]
}
bookflow_db>
```


>_MONGOSH

```
> db.book_metadata.find(
  { $text: { $search: "Digital" } }
).explain("executionStats");
< {
  explainVersion: '1',
  queryPlanner: {
    namespace: 'bookflow_db.book_metadata',
    parsedQuery: {
      '$text': {
        '$search': 'Digital',
        '$language': 'english',
        '$caseSensitive': false,
        '$diacriticSensitive': false
      }
    },
    indexFilterSet: false,
    queryHash: '4FB85230',
    planCacheShapeHash: '4FB85230',
    planCacheKey: 'F34F3876',
    optimizationTimeMillis: 0,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
```

>_MONGOSH

```
winningPlan: {
  isCached: false,
  stage: 'TEXT_MATCH',
  nss: 'bookflow_db.book_metadata',
  indexPrefix: {},
  indexName: 'reviews.comment_text',
  parsedTextQuery: {
    terms: [
      'digit'
    ],
    negatedTerms: [],
    phrases: [],
    negatedPhrases: []
  },
  textIndexVersion: 3,
  inputStage: {
    stage: 'FETCH',
    nss: 'bookflow_db.book_metadata',
    inputStage: {
      stage: 'IXSCAN',
      nss: 'bookflow_db.book_metadata',
      keyPattern: {
        _fts: 'text',
        _ftsx: 1
      },
    },
  },
}
```

>_MONGOSH

```
      indexName: 'reviews.comment_text',
      isMultiKey: true,
      isUnique: false,
      isSparse: false,
      isPartial: false,
      indexVersion: 2,
      direction: 'backward',
      indexBounds: {}
    }
  },
  rejectedPlans: [],
},
executionStats: {
  executionSuccess: true,
  nReturned: 1,
  executionTimeMillis: 1,
  totalKeysExamined: 1,
  totalDocsExamined: 1,
  executionStages: {
    isCached: false,
    stage: 'TEXT_MATCH',
    nReturned: 1,
    executionTimeMillisEstimate: 0,
    works: 2,
```

>_ mongosh: Local Mongo...



>_MONGOSH

```
    advanced: 1,
    needTime: 0,
    needYield: 0,
    saveState: 0,
    restoreState: 0,
    isEOF: 1,
    nss: 'bookflow_db.book_metadata',
    indexPrefix: {},
    indexName: 'reviews.comment_text',
    parsedTextQuery: {
      terms: [
        'digit'
      ],
      negatedTerms: [],
      phrases: [],
      negatedPhrases: []
    },
    textIndexVersion: 3,
    docsRejected: 0,
    inputStage: {
      stage: 'FETCH',
      nReturned: 1,
      executionTimeMillisEstimate: 0,
      works: 2,
      advanced: 1,
      needTime: 0,
```

> mongosh: Local Mongo...



>_MONGOSH

```
    needYield: 0,
    saveState: 0,
    restoreState: 0,
    isEOF: 1,
    nss: 'bookflow_db.book_metadata',
    docsExamined: 1,
    alreadyHasObj: 0,
    inputStage: {
      stage: 'IXSCAN',
      nReturned: 1,
      executionTimeMillisEstimate: 0,
      works: 2,
      advanced: 1,
      needTime: 0,
      needYield: 0,
      saveState: 0,
      restoreState: 0,
      isEOF: 1,
      nss: 'bookflow_db.book_metadata',
      keyPattern: {
        _fts: 'text',
        _ftsx: 1
      },
      indexName: 'reviews.comment_text',
      isMultiKey: true,
      isUnique: false,
```

>_MONGOSH

```
      indexName: 'reviews.comment_text',
      isMultiKey: true,
      isUnique: false,
      isSparse: false,
      isPartial: false,
      indexVersion: 2,
      direction: 'backward',
      indexBounds: {},
      keysExamined: 1,
      seeks: 1,
      dupsTested: 1,
      dupsDropped: 0,
      peakTrackedMemBytes: 8
    }
  }
},
queryShapeHash: '5637A4A384E2ADEAD697F06E65CDEE9F8EE1FEC651F238E25B00AE18EA46711A',
peakTrackedMemBytes: Long('8'),
command: {
  find: 'book_metadata',
  filter: {
    '$text': {
      '$search': 'Digital'
    }
  }
}
```

>_MONGOSH

```
  },
  queryShapeHash: '5637A4A384E2ADEAD697F06E65CDEE9F8EE1FEC651F238E25B00AE18EA46711A',
  peakTrackedMemBytes: Long('8'),
  command: {
    find: 'book_metadata',
    filter: {
      '$text': {
        '$search': 'Digital'
      }
    },
  },
  '$db': 'bookflow_db'
},
serverInfo: {
  host: 'DESKTOP-6S9T6SD',
  port: 27017,
  version: '8.3.8',
  gitVersion: '35e8c8a57f78157ed9fac1a9e90ee6c1818adab6'
},
serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
  internalDocumentSourceGroupMaxMemoryBytes: 104857600,
  internalQueryMaxBlockingSortMemoryUsageBytes: 104857600,
  internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
  internalQueryFacetMaxOutputDocSizeBytes: 104857600,
  internalLookupStageIntermediateDocumentMaxSizeBytes: 104857600,
  internalQueryProhibitBlockingMergeOnMongoS: 0,
```

```
serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
  internalDocumentSourceGroupMaxMemoryBytes: 104857600,
  internalQueryMaxBlockingSortMemoryUsageBytes: 104857600,
  internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
  internalQueryFacetMaxOutputDocSizeBytes: 104857600,
  internalLookupStageIntermediateDocumentMaxSizeBytes: 104857600,
  internalQueryProhibitBlockingMergeOnMongoS: 0,
  internalQueryMaxAddToSetBytes: 104857600,
  internalQueryFrameworkControl: 'trySbeRestricted',
  internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1
}
bookflow_db >
```


>_MONGOSH

```
> db.book_metadata.find(
  { "reviews.comment": { $regex: "Digital", $options: "i" } }
).explain("executionStats");
< {
  explainVersion: '1',
  queryPlanner: {
    namespace: 'bookflow_db.book_metadata',
    parsedQuery: {
      'reviews.comment': BSONRegExp('Digital', 'i')
    },
    indexFilterSet: false,
    queryHash: '819E6067',
    planCacheShapeHash: '819E6067',
    planCacheKey: '29FFD58A',
    optimizationTimeMillis: 0,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: 'COLLSCAN',
      filter: {
        'reviews.comment': BSONRegExp('Digital', 'i')
      },
    },
  },
}
```

>_MONGOSH

```
  nss: 'bookflow_db.book_metadata',
  direction: 'forward'
},
rejectedPlans: []
},
executionStats: {
  executionSuccess: true,
  nReturned: 1,
  executionTimeMillis: 0,
  totalKeysExamined: 0,
  totalDocsExamined: 1,
  executionStages: {
    isCached: false,
    stage: 'COLLSCAN',
    filter: {
      'reviews.comment': BSONRegExp('Digital', 'i')
    },
    nReturned: 1,
    executionTimeMillisEstimate: 0,
    works: 2,
    advanced: 1,
    needTime: 0,
    needYield: 0,
    saveState: 0,
    restoreState: 0,
    isEOF: 1,
```

>_MONGOSH

```
nss: 'bookflow_db.book_metadata',
direction: 'forward',
docsExamined: 1
},
},
queryShapeHash: 'C5E33BF3878BFA00038FB9E1D3EC7AA6170828F1E1E2C026D0DB944D86363DCE',
command: {
  find: 'book_metadata',
  filter: {
    'reviews.comment': BSONRegExp('Digital', 'i')
  },
  '$db': 'bookflow_db'
},
serverInfo: {
  host: 'DESKTOP-6S9T6SD',
  port: 27017,
  version: '8.3.8',
  gitVersion: '35e8c8a57f78157ed9fac1a9e90ee6c1818adab6'
},
serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
  internalDocumentSourceGroupMaxMemoryBytes: 104857600,
  internalQueryMaxBlockingSortMemoryUsageBytes: 104857600,
  internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
  internalQueryFacetMaxOutputDocSizeBytes: 104857600,
  internalLookupStageIntermediateDocumentMaxSizeBytes: 104857600,
```

```
serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
  internalDocumentSourceGroupMaxMemoryBytes: 104857600,
  internalQueryMaxBlockingSortMemoryUsageBytes: 104857600,
  internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
  internalQueryFacetMaxOutputDocSizeBytes: 104857600,
  internalLookupStageIntermediateDocumentMaxSizeBytes: 104857600,
  internalQueryProhibitBlockingMergeOnMongoS: 0,
  internalQueryMaxAddToSetBytes: 104857600,
  internalQueryFrameworkControl: 'trySbeRestricted',
  internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1
}
bookflow_db > |
```

```
>_MONGOSH
```

```
> db.book_metadata.createIndex({ formats: 1 });
< formats_1
> db.book_metadata.createIndex({ avgRating: -1 });
< avgRating_-1
> db.book_metadata.find(
  { avgRating: { $gt: 4 } }
).explain("executionStats");
< {
  explainVersion: '1',
  queryPlanner: {
    namespace: 'bookflow_db.book_metadata',
    parsedQuery: {
      avgRating: {
        '$gt': 4
      }
    },
    indexFilterSet: false,
    queryHash: 'E21FCB34',
    planCacheShapeHash: 'E21FCB34',
    planCacheKey: '6A4442CC',
    optimizationTimeMillis: 0,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
```

>_MONGOSH

```
winningPlan: {
  isCached: false,
  stage: 'FETCH',
  nss: 'bookflow_db.book_metadata',
  inputStage: {
    stage: 'IXSCAN',
    nss: 'bookflow_db.book_metadata',
    keyPattern: {
      avgRating: -1
    },
    indexName: 'avgRating_-1',
    isMultiKey: false,
    multiKeyPaths: {
      avgRating: []
    },
    isUnique: false,
    isSparse: false,
    isPartial: false,
    indexVersion: 2,
    direction: 'forward',
    indexBounds: {
      avgRating: [
        '[inf, 4)'
      ]
    }
  }
}
```

>_MONGOSH

```
  },
  rejectedPlans: [],
},
executionStats: {
  executionSuccess: true,
  nReturned: 1,
  executionTimeMillis: 4,
  totalKeysExamined: 1,
  totalDocsExamined: 1,
  executionStages: {
    isCached: false,
    stage: 'FETCH',
    nReturned: 1,
    executionTimeMillisEstimate: 0,
    works: 2,
    advanced: 1,
    needTime: 0,
    needYield: 0,
    saveState: 0,
    restoreState: 0,
    isEOF: 1,
    nss: 'bookflow_db.book_metadata',
    docsExamined: 1,
    alreadyHasObj: 0,
    inputStage: {
      stage: 'IXSCAN',
```

>_MONGOSH

```
nReturned: 1,
executionTimeMillisEstimate: 0,
works: 2,
advanced: 1,
needTime: 0,
needYield: 0,
saveState: 0,
restoreState: 0,
isEOF: 1,
nss: 'bookflow_db.book_metadata',
keyPattern: {
  avgRating: -1
},
indexName: 'avgRating_-1',
isMultiKey: false,
multiKeyPaths: {
  avgRating: []
},
isUnique: false,
isSparse: false,
isPartial: false,
indexVersion: 2,
direction: 'forward',
indexBounds: {
  avgRating: [
    '[inf, 4)'
  ]
}
```

>_MONGOSH

```
    ]
  },
  keysExamined: 1,
  seeks: 1,
  dupsTested: 0,
  dupsDropped: 0,
  peakTrackedMemBytes: 0
}
},
queryShapeHash: '7FF7F0A0735518208765478E686FE29C704C6FEFB53464ADDD51CDD797ACB93',
command: {
  find: 'book_metadata',
  filter: {
    avgRating: {
      '$gt': 4
    }
  },
  '$db': 'bookflow_db'
},
serverInfo: {
  host: 'DESKTOP-6S9T6SD',
  port: 27017,
  version: '8.3.8',
  gitVersion: '35e8c8a57f78157ed9fac1a9e90ee6c1818adab6'
},
```



```
serverInfo: {
  host: 'DESKTOP-6S9T6SD',
  port: 27017,
  version: '8.3.8',
  gitVersion: '35e8c8a57f78157ed9fac1a9e90ee6c1818adab6'
},
serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
  internalDocumentSourceGroupMaxMemoryBytes: 104857600,
  internalQueryMaxBlockingSortMemoryUsageBytes: 104857600,
  internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
  internalQueryFacetMaxOutputDocSizeBytes: 104857600,
  internalLookupStageIntermediateDocumentMaxSizeBytes: 104857600,
  internalQueryProhibitBlockingMergeOnMongoS: 0,
  internalQueryMaxAddToSetBytes: 104857600,
  internalQueryFrameworkControl: 'trySbeRestricted',
  internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1
}
bookflow_db >
```

```
>_MONGOSH
```

```
> db.book_metadata.find(
  { formats: "Digital" }
).explain("executionStats");
< {
  explainVersion: '1',
  queryPlanner: {
    namespace: 'bookflow_db.book_metadata',
    parsedQuery: {
      formats: {
        '$eq': 'Digital'
      }
    },
    indexFilterSet: false,
    queryHash: '2E656ACA',
    planCacheShapeHash: '2E656ACA',
    planCacheKey: 'E7E31F30',
    optimizationTimeMillis: 0,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: 'FETCH',
      nss: 'bookflow_db.book_metadata',
      inputStage: {
```

>_MONGOSH

```
stage: 'FETCH',
nss: 'bookflow_db.book_metadata',
inputStage: {
  stage: 'IXSCAN',
  nss: 'bookflow_db.book_metadata',
  keyPattern: {
    formats: 1
  },
  indexName: 'formats_1',
  isMultiKey: true,
  multiKeyPaths: {
    formats: [
      'formats'
    ]
  },
  isUnique: false,
  isSparse: false,
  isPartial: false,
  indexVersion: 2,
  direction: 'forward',
  indexBounds: {
    formats: [
      '["Digital", "Digital"]'
    ]
  }
}
```

>_MONGOSH

```
    },
    rejectedPlans: [],
  },
  executionStats: {
    executionSuccess: true,
    nReturned: 1,
    executionTimeMillis: 0,
    totalKeysExamined: 1,
    totalDocsExamined: 1,
    executionStages: {
      isCached: false,
      stage: 'FETCH',
      nReturned: 1,
      executionTimeMillisEstimate: 0,
      works: 2,
      advanced: 1,
      needTime: 0,
      needYield: 0,
      saveState: 0,
      restoreState: 0,
      isEOF: 1,
      nss: 'bookflow_db.book_metadata',
      docsExamined: 1,
      alreadyHasObj: 0,
      inputStage: {
        stage: 'IXSCAN',
```

>_MONGOSH

```
nReturned: 1,
executionTimeMillisEstimate: 0,
works: 2,
advanced: 1,
needTime: 0,
needYield: 0,
saveState: 0,
restoreState: 0,
isEOF: 1,
nss: 'bookflow_db.book_metadata',
keyPattern: {
  formats: 1
},
indexName: 'formats_1',
isMultiKey: true,
multiKeyPaths: {
  formats: [
    'formats'
  ]
},
isUnique: false,
isSparse: false,
isPartial: false,
indexVersion: 2,
direction: 'forward',
indexBounds: {
```

>_MONGOSH

```
    formats: [
      ["Digital", "Digital"]
    ],
    keysExamined: 1,
    seeks: 1,
    dupsTested: 1,
    dupsDropped: 0,
    peakTrackedMemBytes: 8
  }
},
queryShapeHash: '7284F4601D06C908F75AA2060148D728A34169E5F97317861278D07F8D685B33',
peakTrackedMemBytes: Long('8'),
command: {
  find: 'book_metadata',
  filter: {
    formats: 'Digital'
  },
  '$db': 'bookflow_db'
},
serverInfo: {
  host: 'DESKTOP-6S9T6SD',
  port: 27017,
  version: '8.3.8',
  gitVersion: '35e8c8a57f78157ed9fac1a9e90ee6c1818adab6'
```

```
},
serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
  internalDocumentSourceGroupMaxMemoryBytes: 104857600,
  internalQueryMaxBlockingSortMemoryUsageBytes: 104857600,
  internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
  internalQueryFacetMaxOutputDocSizeBytes: 104857600,
  internalLookupStageIntermediateDocumentMaxSizeBytes: 104857600,
  internalQueryProhibitBlockingMergeOnMongoS: 0,
  internalQueryMaxAddToSetBytes: 104857600,
  internalQueryFrameworkControl: 'trySbeRestricted',
  internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1
}
bookflow_db >
```

>_MONGOSH

```
> db.book_metadata.find(
  { "reviews.comment": { $regex: "Digital", $options: "i" } }
).explain("executionStats");
< {
  explainVersion: '1',
  queryPlanner: {
    namespace: 'bookflow_db.book_metadata',
    parsedQuery: {
      'reviews.comment': BSONRegExp('Digital', 'i')
    },
    indexFilterSet: false,
    queryHash: '819E6067',
    planCacheShapeHash: '819E6067',
    planCacheKey: '29FFD58A',
    optimizationTimeMillis: 0,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: 'COLLSCAN',
      filter: {
        'reviews.comment': BSONRegExp('Digital', 'i')
      },
    },
  },
}
```

>_MONGOSH

```
    nss: 'bookflow_db.book_metadata',
    direction: 'forward'
  },
  rejectedPlans: []
},
executionStats: {
  executionSuccess: true,
  nReturned: 1,
  executionTimeMillis: 1,
  totalKeysExamined: 0,
  totalDocsExamined: 1,
  executionStages: {
    isCached: false,
    stage: 'COLLSCAN',
    filter: {
      'reviews.comment': BSONRegExp('Digital', 'i')
    },
    nReturned: 1,
    executionTimeMillisEstimate: 0,
    works: 2,
    advanced: 1,
    needTime: 0,
    needYield: 0,
    saveState: 0,
    restoreState: 0,
```


>_MONGOSH

```
  nReturned: 1,
  executionTimeMillisEstimate: 0,
  works: 2,
  advanced: 1,
  needTime: 0,
  needYield: 0,
  saveState: 0,
  restoreState: 0,
  isEOF: 1,
  nss: 'bookflow_db.book_metadata',
  direction: 'forward',
  docsExamined: 1
}
},
queryShapeHash: 'C5E33BF3878BFA00038FB9E1D3EC7AA6170828F1E1E2C026D0DB944D86363DCE',
command: {
  find: 'book_metadata',
  filter: {
    'reviews.comment': BSONRegExp('Digital', 'i')
  },
  '$db': 'bookflow_db'
},
serverInfo: {
  host: 'DESKTOP-6S9T6SD',
  port: 27017,
  version: '8.3.8',
```

```
serverInfo: {
  host: 'DESKTOP-6S9T6SD',
  port: 27017,
  version: '8.3.8',
  gitVersion: '35e8c8a57f78157ed9fac1a9e90ee6c1818adab6'
},
serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
  internalDocumentSourceGroupMaxMemoryBytes: 104857600,
  internalQueryMaxBlockingSortMemoryUsageBytes: 104857600,
  internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
  internalQueryFacetMaxOutputDocSizeBytes: 104857600,
  internalLookupStageIntermediateDocumentMaxSizeBytes: 104857600,
  internalQueryProhibitBlockingMergeOnMongoS: 0,
  internalQueryMaxAddToSetBytes: 104857600,
  internalQueryFrameworkControl: 'trySbeRestricted',
  internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1
}
bookflow_db > |
```

>_MONGOSH

```
> db.book_metadata.createIndex({ formats: 1 });
< formats_1
> db.book_metadata.createIndex({ avgRating: -1 });
< avgRating_-1
> db.book_metadata.find(
  { avgRating: { $gt: 4 } }
).explain("executionStats");
< {
  explainVersion: '1',
  queryPlanner: {
    namespace: 'bookflow_db.book_metadata',
    parsedQuery: {
      avgRating: {
        '$gt': 4
      }
    },
    indexFilterSet: false,
    queryHash: 'E21FCB34',
    planCacheShapeHash: 'E21FCB34',
    planCacheKey: '6A4442CC',
    optimizationTimeMillis: 0,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
```

>_MONGOSH

```
winningPlan: {
  isCached: false,
  stage: 'FETCH',
  nss: 'bookflow_db.book_metadata',
  inputStage: {
    stage: 'IXSCAN',
    nss: 'bookflow_db.book_metadata',
    keyPattern: {
      avgRating: -1
    },
    indexName: 'avgRating_-1',
    isMultiKey: false,
    multiKeyPaths: {
      avgRating: []
    },
    isUnique: false,
    isSparse: false,
    isPartial: false,
    indexVersion: 2,
    direction: 'forward',
    indexBounds: {
      avgRating: [
        '[inf, 4)'
      ]
    }
  }
}
```

>_MONGOSH

```
    rejectedPlans: []
  },
  executionStats: {
    executionSuccess: true,
    nReturned: 1,
    executionTimeMillis: 0,
    totalKeysExamined: 1,
    totalDocsExamined: 1,
    executionStages: {
      isCached: false,
      stage: 'FETCH',
      nReturned: 1,
      executionTimeMillisEstimate: 0,
      works: 2,
      advanced: 1,
      needTime: 0,
      needYield: 0,
      saveState: 0,
      restoreState: 0,
      isEOF: 1,
      nss: 'bookflow_db.book_metadata',
      docsExamined: 1,
      alreadyHasObj: 0,
      inputStage: {
        stage: 'IXSCAN',
```

>_MONGOSH

```
nReturned: 1,
executionTimeMillisEstimate: 0,
works: 2,
advanced: 1,
needTime: 0,
needYield: 0,
saveState: 0,
restoreState: 0,
isEOF: 1,
nss: 'bookflow_db.book_metadata',
keyPattern: {
  avgRating: -1
},
indexName: 'avgRating_-1',
isMultiKey: false,
multiKeyPaths: {
  avgRating: []
},
isUnique: false,
isSparse: false,
isPartial: false,
indexVersion: 2,
direction: 'forward',
indexBounds: {
  avgRating: [
    '[inf, 4)'
  ]
}
```

>_MONGOSH

```
    ]
  },
  keysExamined: 1,
  seeks: 1,
  dupsTested: 0,
  dupsDropped: 0,
  peakTrackedMemBytes: 0
}
},
queryShapeHash: '7FF7F0A0735518208765478E686FE29C704C6FEFB53464ADDD51CDD797ACB93',
command: {
  find: 'book_metadata',
  filter: {
    avgRating: {
      '$gt': 4
    }
  },
  '$db': 'bookflow_db'
},
serverInfo: {
  host: 'DESKTOP-6S9T6SD',
  port: 27017,
  version: '8.3.8',
  gitVersion: '35e8c8a57f78157ed9fac1a9e90ee6c1818adab6'
},
```

```
},
serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
  internalDocumentSourceGroupMaxMemoryBytes: 104857600,
  internalQueryMaxBlockingSortMemoryUsageBytes: 104857600,
  internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
  internalQueryFacetMaxOutputDocSizeBytes: 104857600,
  internalLookupStageIntermediateDocumentMaxSizeBytes: 104857600,
  internalQueryProhibitBlockingMergeOnMongoS: 0,
  internalQueryMaxAddToSetBytes: 104857600,
  internalQueryFrameworkControl: 'trySbeRestricted',
  internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1
}
bookflow_db >
```

